4. **Separation of DNA by electrophoresis**
During electrophoresis in an agarose gel, the DNA fragments are separated into bands of different sizes. The bands of separated DNA are sieved out of the gel using a nylon membrane (treated with chemicals that allow for it to break the hydrogen bonds of DNA so there are single strands).
5. **Denaturing DNA**
The DNA on gels is denatured by using alkaline chemicals or by heating.
6. **Blotting**
The DNA band pattern in the gel is transferred to a thin nylon membrane placed over the 'size fractionated DNA strand' by **Southern blotting**.
7. **Using probes to identify specific DNA**
A radioactive probe (DNA labeled with a radioactive substance) is added to the DNA bands. The probe attaches by base pairing to those restriction fragments that are complementary to its sequence. The probes can also be prepared by using either 'fluorescent substance' or 'radioactive isotopes'.
8. **Hybridization with probe**
After the probe hybridizes and the excess probe washed off, a photographic film is placed on the membrane containing 'DNA hybrids'.
9. **Exposure on film to make a genetic/DNA Fingerprint**
The radioactive label exposes the film to form an image (image of bands) corresponding to specific DNA bands. The thick and thin dark bands form a pattern of **bars** which constitutes a genetic fingerprint.

Application of DNA finger printing

- **Forensic analysis** - It can be used in the identification of a person involved in criminal activities, for settling paternity or maternity disputes, and in determining relationships for immigration purposes.
- **Pedigree analysis** – inheritance pattern of genes through generations and for detecting inherited diseases.
- **Conservation of wild life** – protection of endangered species. By maintaining DNA records for identification of tissues of the dead endangered organisms.
- **Anthropological studies**–It is useful in determining the origin and migration of human populations and genetic diversities.

Summary

In the twentieth century, one of the landmark discovery in biology was the identification of DNA, as genetic material of living organisms. Gene may be defined as a segment of DNA which is responsible for inheritance and expression of a particular character.

In 1953, James Watson and Francis Crick proposed DNA structure based on X-ray crystallographic studies provided by Maurice Wilkins and Rosalind Franklin. Nucleotides are the structural units of nucleic acids. Each nucleotide has three components, i) pentose sugar ii) nitrogenous base and iii) phosphate. DNA and RNA are polynucleotides. DNA has double stranded helical structure while RNA is a single stranded structure. DNA acts as genetic material of almost all the living organism except few viruses.

The non genetic RNAs are of three types; m-RNA, r-RNA and t-RNA. They help in protein synthesis. DNA has capacity of replication, while the three types of RNA are transcribed on DNA. Meselson and Stahl (1958) proved experimentally the semi-conservative nature of DNA replication using heavy isotope of nitrogen N^{15} in *E.coli*.

In 1958 Crick proposed that DNA determines the sequence of amino acids in a polypeptide (protein) through mRNA, and proposed the central dogma of protein synthesis which involves transcription and translation. The process of copying genetic information from one strand of DNA into RNA is termed transcription. The DNA transcribed RNA molecules serve as a template for the synthesis of polypeptides by a process termed translation. Each amino acid in

TN_GOVT_XII_Zoology_chapter5.indd 86 24-02-2020 4.01.39 PM